Entropy Stable Galerkin Projection Reduced Order Model for the Discontinuous Galerkin Method

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CSME - CFDSC - CSR 2025
May 26, 2025

Outline

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1. Entropy Stable Discontinuous Galerkin
2. Reduced Order Modeling
3. Hyper Reduction
4. Entropy Stable Reduced Order Model
5. Entropy Stable Hyper Reduction
6. Numerical Results

Introduction

- Computational models have been increasingly essential in design and engineering
- However, high costs and storage requirements lead to bottlenecks

$\longrightarrow$ Goal: Reduce the size of the problem

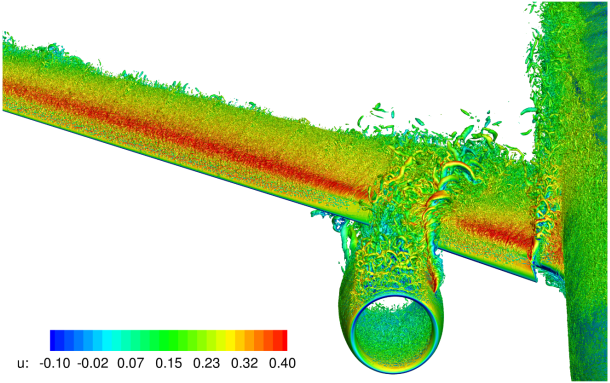


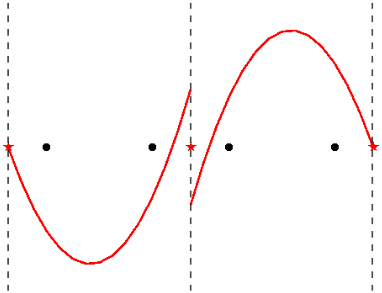
Figure: NASA high-lift common research model at $\alpha = 21.47^o$, $M = 0.2$, $Re = 5.49 \times 10^6$, simulated using WMLES.[1]

[1] Wang, Li et al. *AIAA SciTech 2022 Forum*. 2022.

Discontinuous Galerkin

The discontinuous Galerkin (DG) method[2]

- Represents the solution as high degree polynomials in elements



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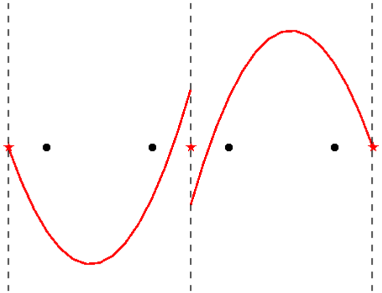

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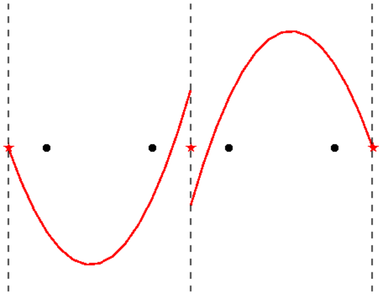
Discontinuous Galerkin

The discontinuous Galerkin (DG) method[2]

- Represents the solution as high degree polynomials in elements
- Allows the solution to be discontinuous at element interfaces
- Ensures the conservation of variables among elements



$\longrightarrow$ Higher resolution per degree of freedom than low-order methods.

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- Converge to non-physical solutions
- Non-linear phenomena are unstable, potentially crashing the simulation

This motivates the need for entropy stability

$$\text{DG Split Form Method: } \mathbf{M}\frac{d\mathbf{u}_h}{dt} + (\mathbf{Q} \circ \mathbf{F})\mathbf{1} = \mathbf{0}$$

Entropy Stable Discontinuous Galerkin

For a DG method to be entropy stable, its skew symmetric operator, $\mathbf{Q}$, has the following properties[3]:

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There is one additional relevant condition to ensure general entropy stability for DG:

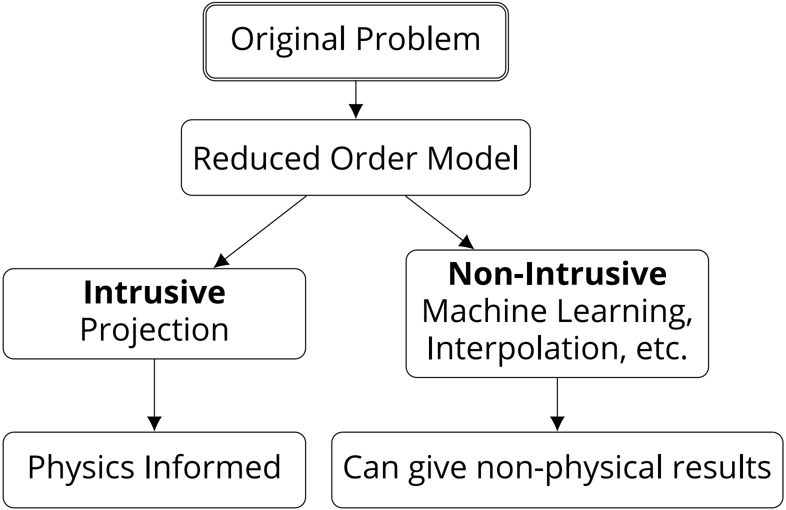
- The flux must be calculated using projected entropy variables, with the following projection operator, $\mathbf{\Pi}_V$

[3] Chan, Jesse. *Journal of Scientific Computing*. 2019.

Reduced Order Modeling

To reduce the size of the problem, we use a reduced order model (ROM).

- Approximates the behavior of the original high-fidelity model



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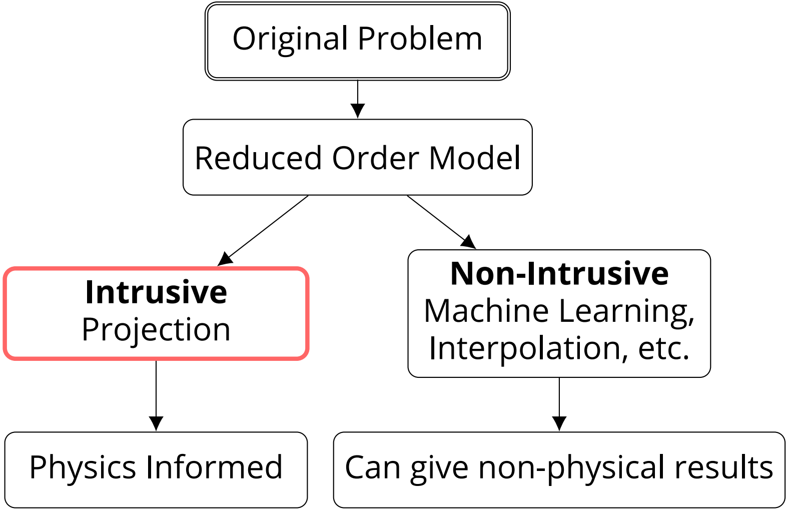

Projection Based Reduced Order Modeling

We assume that the solution belongs on a lower-dimension subspace, $\mathbf{V} \in \mathbb{R}^{N \times N_R}$, a trial basis

$$\mathbf{u}_h = \mathbf{u}_{ref} + \mathbf{V}\mathbf{u}_r$$

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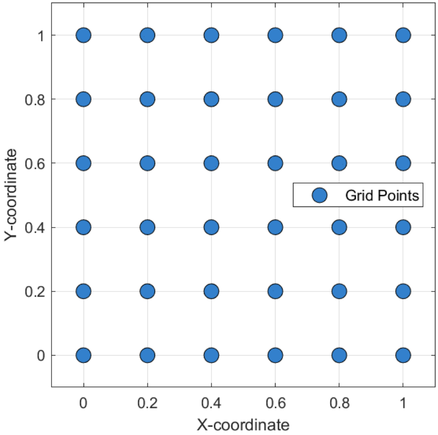
$$N \times N \longrightarrow N_R \times N_R, \quad N_R << N$$

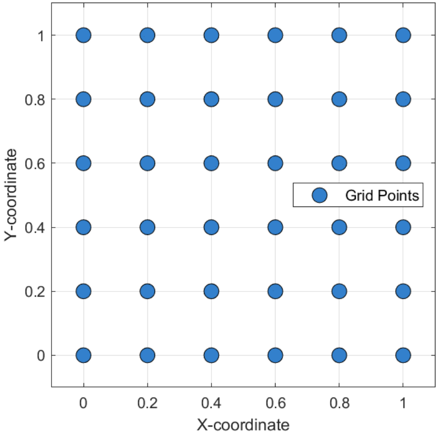
Hyper Reduction

- Quadrature points are sub selected into a the hyper reduced set



Hyper Reduction

- Quadrature points are sub selected into a the hyper reduced set
- Quadrature weighting are adjusted to account for importance



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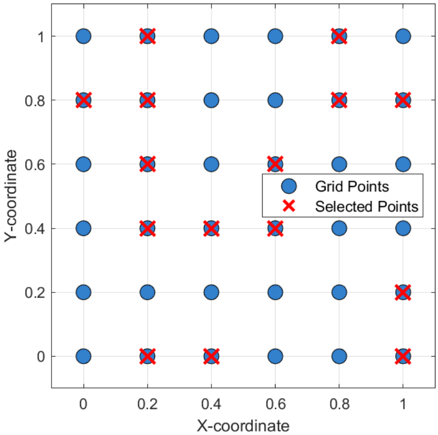

$\longrightarrow$ With hyper reduction, **Q** loses its stability properties

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- In a ROM, the projection preserves the conservation laws

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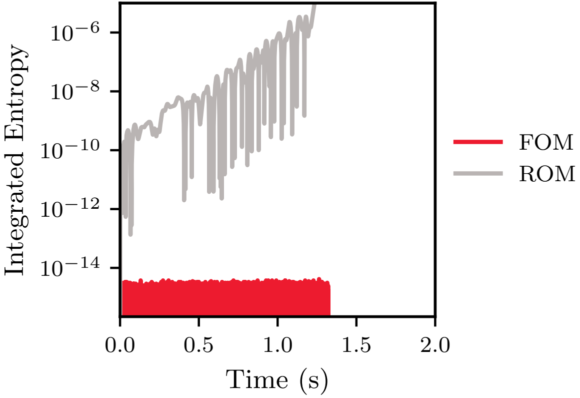

Figure: ROM Integrated Entropy of an Euler problem

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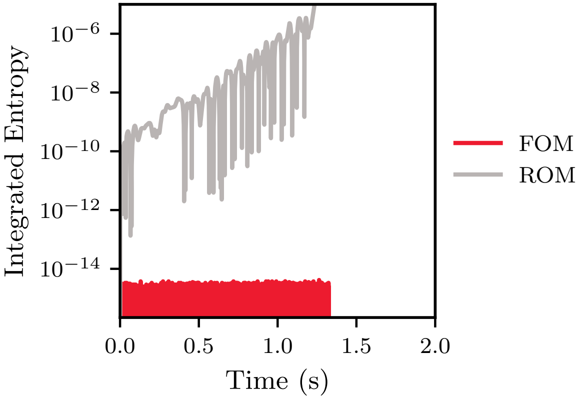

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$\longrightarrow$ To ensure entropy stability, we project the entropy variables into the reduced order space

Hyper Reduction: Preserving Entropy Stability

- To preserve the entropy stability properties, we sacrifice the sparsity of the problem

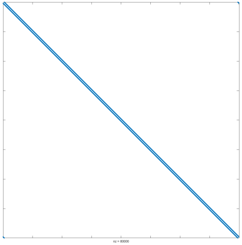
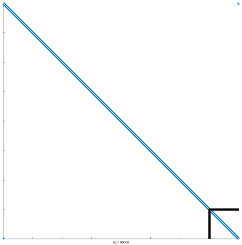


Figure: Sparsity Pattern of Sparse $N \times N$ Matrix

Hyper Reduction: Preserving Entropy Stability

- To preserve the entropy stability properties, we sacrifice the sparsity of the problem
- The **Q** operator is projected into a hyper reduced space to create a **dense** matrix:



$\longrightarrow$

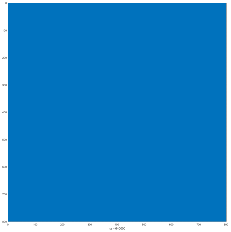


Figure: Sparsity Pattern of Sparse $N \times N$ Matrix

Figure: Sparsity Pattern of Dense $N_H \times N_H$ Matrix

1D Boundary Conditions

- The global boundary integration matrix, **B**, stores the normals

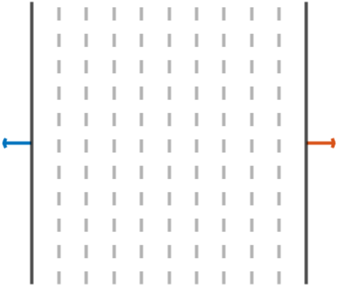


Figure: Global Normal Vectors of 1D Problem

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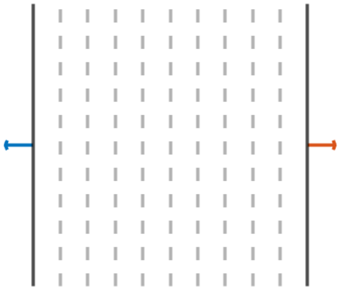

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- The global extrapolation matrix,**E**, is projected into the reduced order space and then projected back into the hyper reduced space

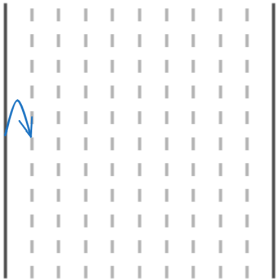

Figure: Full Order Extrapolation Matrix

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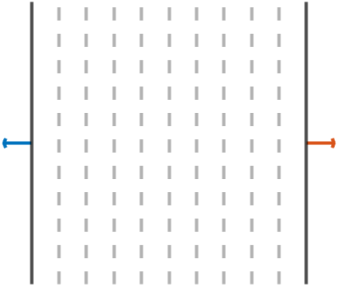

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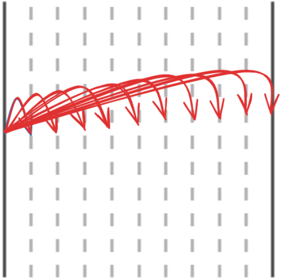

Figure: Hyper Reduced Extrapolation Matrix

Cost Estimation

Online Costs:

- Flux calculations of efficient DG code: $O(N)$
- Flux calculations of HESROM: $O(N_H^2)$

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Offline costs:

- Construction of basis, $\mathbf{V}$
- Calculation of hyper reduction
- Construction of projection matrix, $\mathbf{\Pi}$

Test Case: 1D Shock

This 1D shock[4] contains a wall boundary condition on either side of the domain

- c_{Hu} Flux Reconstruction[5]
- Lax-Friedrichs Dissipation
- Chandrashekar Flux with Ranocha fix
- 3 Stage Explicit SSPRK
- 6th Order Polynomial
- GL Solution Nodes
- Starting CFL = 0.1

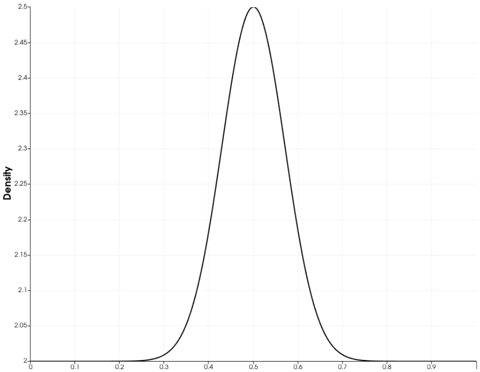

Figure: Density along Domain

[4] Chan, Jesse. *Journal of Computational Physics*. 2020.
[5] Huynh, Hung T. 2007.

1D Shock: Singular Values

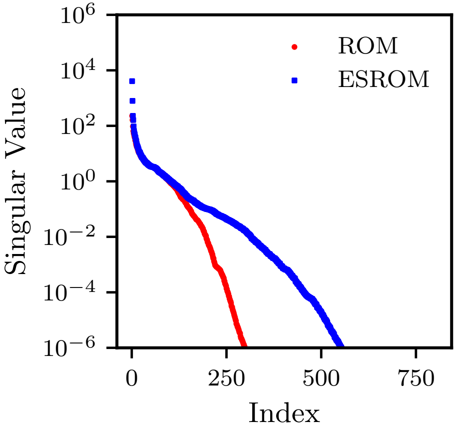

Figure: Singular Values, p2, CFL 0.35, c+, 1D Shock

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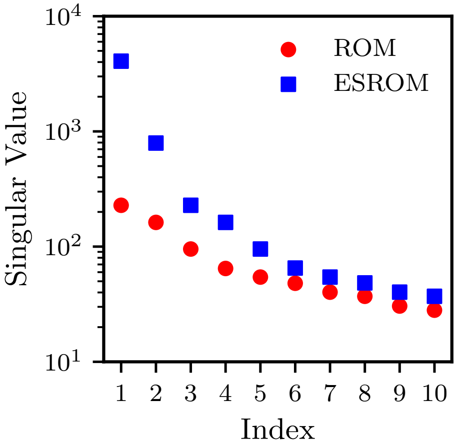


Figure: Singular Values, p2, CFL 0.35, c+, 1D Shock Zoomed in

1D Shock: Increasing Polynomial Order

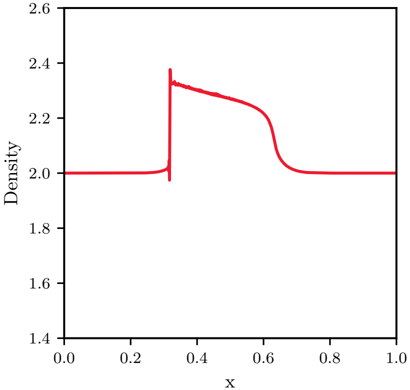


Figure: Density of p6, CFL 0.1, c_{Hu} 1D at $t = 0.75$s

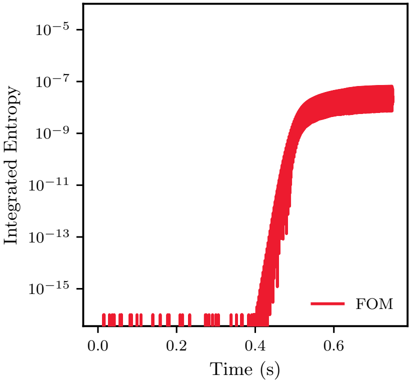
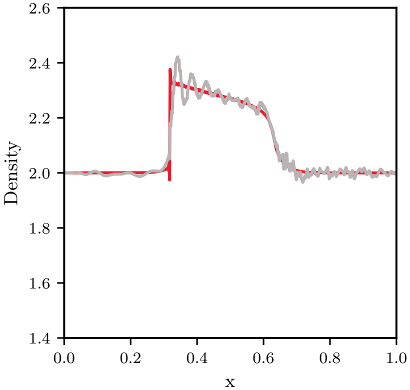


Figure: Integrated Entropy of p6, CFL 0.1, c_{Hu}, 1D Shock



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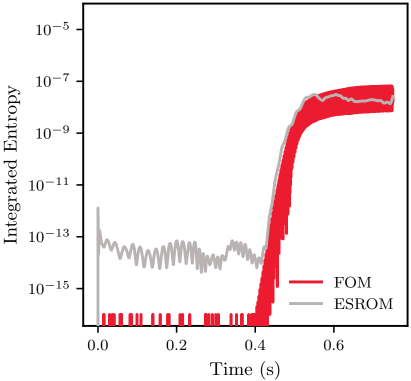

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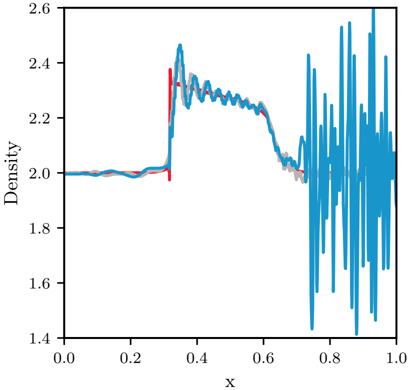

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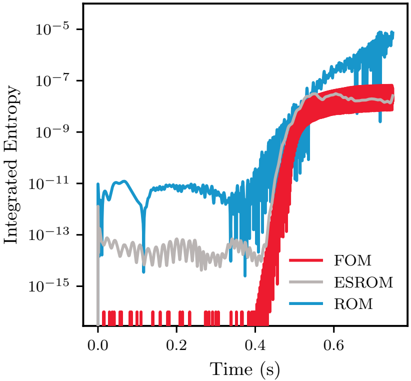

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1D Shock: Hyper Reduction

$$p = 2, c = c_{DG}, \text{ CFL} = 0.5$$

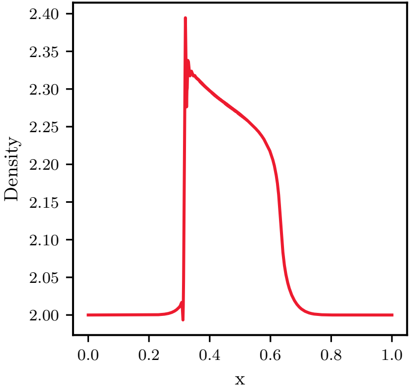
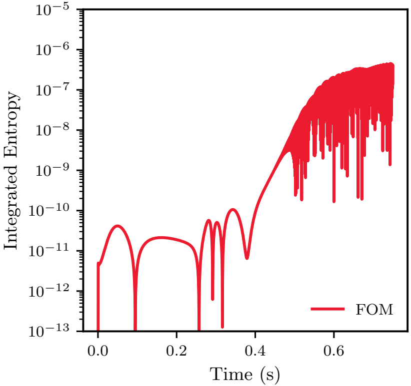


Figure: Density of p2, CFL 0.5, c_{DG} 1D at $t = 0.75$s



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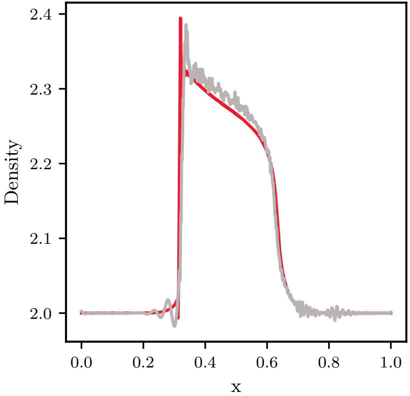

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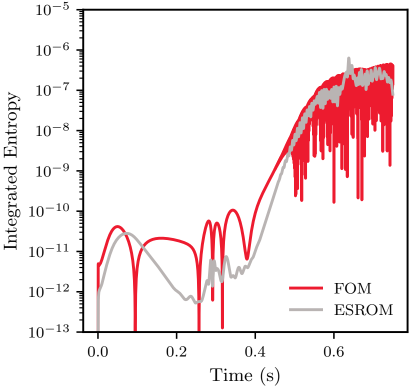

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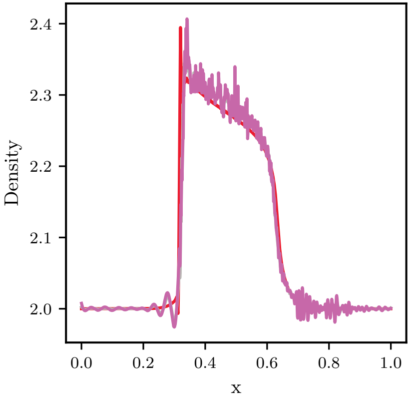

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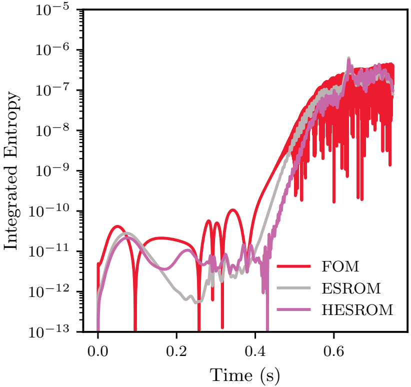

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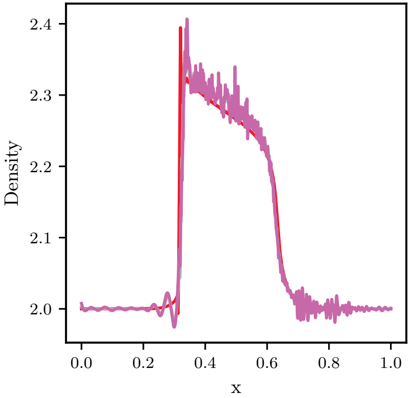

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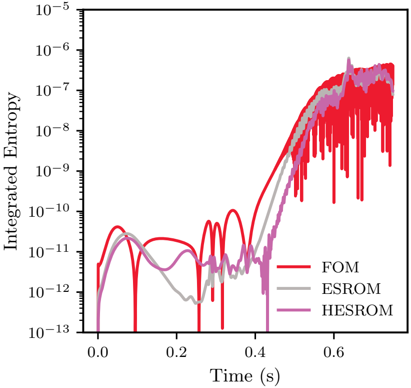

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4608 DoFs $\longrightarrow$ 2805 Hyper DoFs, 800 reduced order variables

Isentropic Vortex

To test for the models performance on a variety of grid sizes, we test an isentropic vortex advection[6] across $x = y$

- Chandrashekar Flux with Ranocha fix
- GLL Solution Nodes

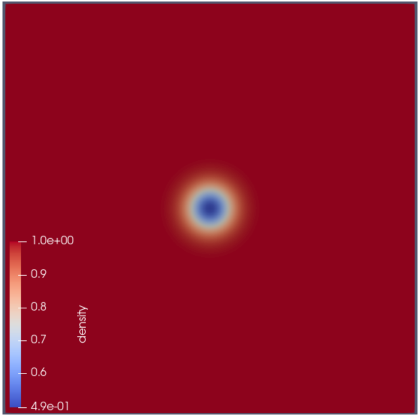

Figure: Isentropic vortex advection on a $[-10,10]^2$ domain.

[6] Wang, Zhijian J et al. *International Journal for Numerical Methods in Fluids*. 2013.

Isentropic Vortex: Grid Refinement

Convergence order of the fully-discrete scheme while refining an $n \times n$ grid:

Scheme	n	L^2 error	L^2 rate	ESROM L^2 error	ESROM L^2 rate
SSPRK3, $p = 2$	8	6.532E-1	-	6.531E-1	-
	16	3.885E-1	0.75	3.888E-1	0.75
	32	3.889E-2	3.32	3.890E-2	3.32
	64	6.309E-3	2.62	6.303E-3	2.63
	128	8.006E-4	**2.98**	8.106E-4	**2.96**
RK4, $p = 3$	8	4.009E-1	-	4.009E-1	-
	16	3.464E-2	3.53	3.432E-2	3.53
	32	3.217E-3	3.43	3.227E-3	3.43
	64	3.616E-4	**3.15**	3.731E-4	**3.11**

Isentropic Vortex: Integrated Entropy

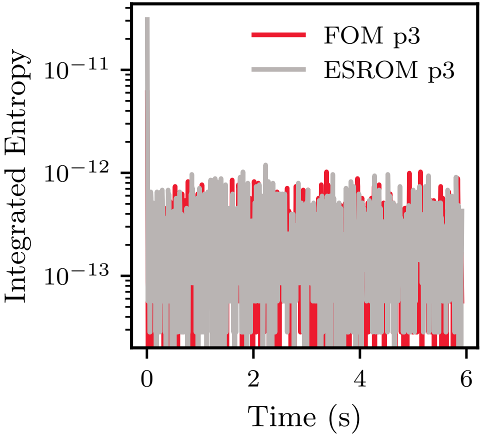

Figure: Integrated Entropy for $p3$ Vortex Advection 64^2 cells

Conclusions and Future Work

Conclusion

- Entropy stable ROMs track the full order solution and remain entropy stable for the length of the simulation
- The hyper reduced entropy stable ROM has been developed for multiple dimensions and different boundary conditions

Future Work

- Reduce the offline costs
- Optimization of the flux calculations

Acknowledgements

We are grateful for our sources of funding, Bombardier and **McGill University**.

All test cases use the `PHiLiP` library (Shi-Dong and Nadarajah, 2021). Some results use the Narval compute cluster, supported by **Calcul Québec** and the **Digital Research Alliance of Canada**.

Questions?

Thank you for your time!